Chapter 1 Turning Effect of Forces

Objectives

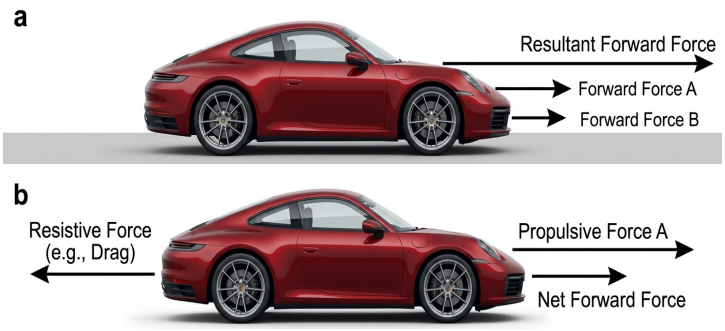
You will be able to:

-Add forces.
-Describe how to resolve force into components.
-Understand the turning effect of forces.
-Calculate moment.
-Apply the principle of moment.
-Add parallel forces.
-Find the center of gravity.

1-Adding forces in the same line

Forces are vectors. This means that measuring their magnitude is important but equally important is knowing the direction in which they act. In order to calculate the overall effect of multiple forces acting on the same object, we can use vector addition to work out the resultant force. This resultant force can be considered as a single fore that has the same effect as all the individual forces combined.

If two or more forces are acting along the same line, then combining them is simply a case of adding or subtracting their magnitudes depending on their directions.

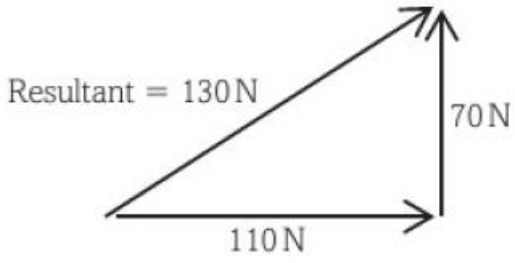


Adding perpendicular forces

The effect on an object of two forces that are acting at right angles (perpendicular) to each other will be the vector sum of their individual effects. We need to add the sizes with consideration for the directions in order to find the resultant

Magnitude of the resultant force

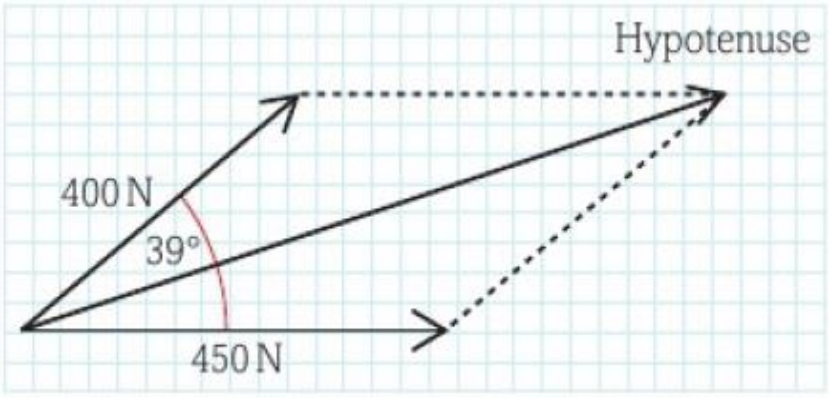
To calculate the resultant magnitude of two perpendicular forces, we can draw them, one after the other, as the two sides of a right-angled triangle and use Pythagoras' theorem to calculate the size of the hypotenuse.



The resultant force here is calculated using Pythagoras' theorem:
$F = \sqrt{(70^2 + 110^2)} = 130\,N$

The parallelogram rule

There is another method for finding the resultant of non-perpendicular foxes (or vectors) by scale drawing, which can be considered to use. This is called the parallelogram rule. Draw the two vectors to scale — at the correct angle and scaled so their length represents the magnitude — starting from the same point, Then draw the same two vectors again parallel to the original ones, so that they form a parallelogram, as shown in fig. The resultant fore (or vector) will be the diagonal across the parallelogram from the starting point.



Finding the resultant vector using the parallelogram rule.

Questions:

1.Work out the resultant force on a toy car if it has the following forces acting on it:

-A rubber band motor driving forwards 8.4N.

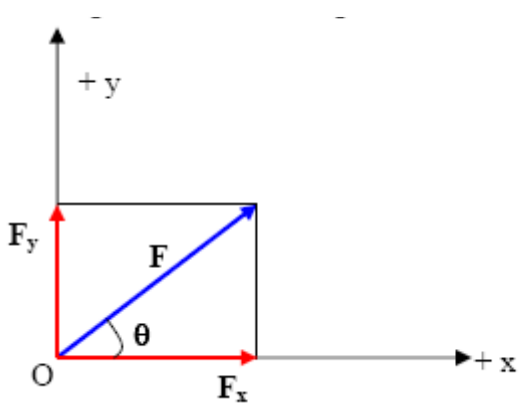
-An air resistance 0.5 N

-A friction 5.8 N

-A child hand pushing forward 10 N.

2. As a small plane takes off, the lift force on it is 6000 N vertically upwards, while the thrust is 2800 N horizontally forwards. What is the resultant of these forces on the plane?.

2-Resolving Force



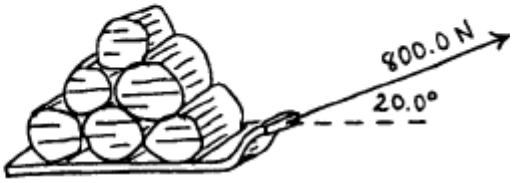
It is always possible to resolve a force into two components along any given pair of perpendicular directions. In order to resolve a force into its own components we need a suitable **coordinate** system. We generally define the horizontal axis as x-axis and vertical axis as y-axis. Any given force F can be drawn with the proper angle α such that its tail coincides with the origin of the axes as shown. The components along the x- and y-axes of this force are usually called F_x and F_y respectively. To find these components draw perpendicular lines from the head of r to x and y axes. The magnitudes of these components are then found using the trigonometric equations.

F_x, F_y are called x-y component of F.

$F_x = F\cos\theta\ , F_y = F\sin\theta$

Practice Exercises

1-Mark pulls a sled loaded with logs to his cabin in the woods. I Mark pull with a force of 800N in a direction of 20^0 above the horizontal, what are the horizontal and vertical components of the force exerted by Mark?



2-Marcie shovels snow after a storm by exerting a force of 30N on her shovel at an angle of 60^0 to the vertical. What are the horizontal and vertical components of the force exerted by Marcie?

3-A man pulls a sled by exerting a force of 35 N on a rope making an angle of 55° with the horizontal. Find the vertical and horizontal components of the force.

4-A Person pushes a lawn mower by exerting a 76 N force on the handle inclined at 40.0° above the horizontal.

a. What force pushes the mower against the lawn (the vertical component)?

b. What force moves it forward (the horizontal component)?

5-Find the x and y components (i.e. the horizontal and vertical components) of an 88 N force making an angle of 22° with the x-axis.

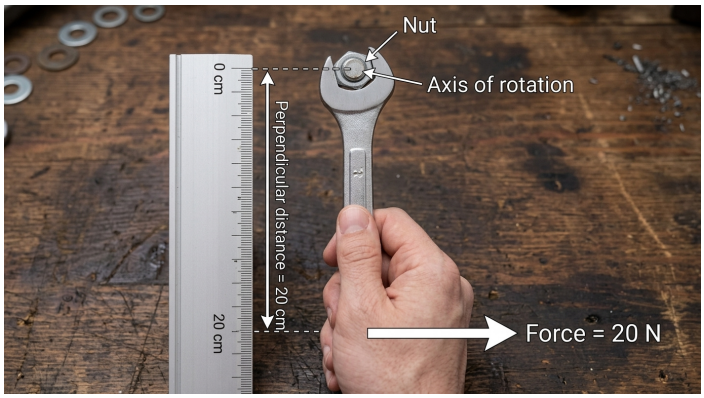
3-Turning Effect of Force

It is easier to loosen a nut with a long spanner than with a short one.

Or, it is easier to loosen a nut with a larger force than with a weak force.

The turning effect of a force is called the **moment of the force**. It depends on both the size of the force and how far it is applied from the pivot or fulcrum. It is measured by multiplying the force by the perpendicular distance of the line of

action of the force from the fulcrum. The unit is the newton meter (N.m).

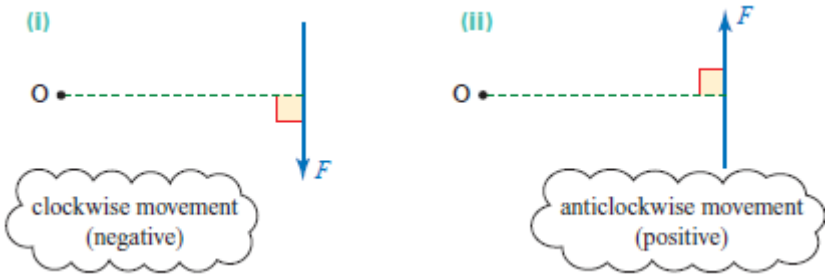


The moment of a force (or torque) is the product of the force(F) and the perpendicular distance (d) from the pivot to the line of action of the force.

M=Fxd

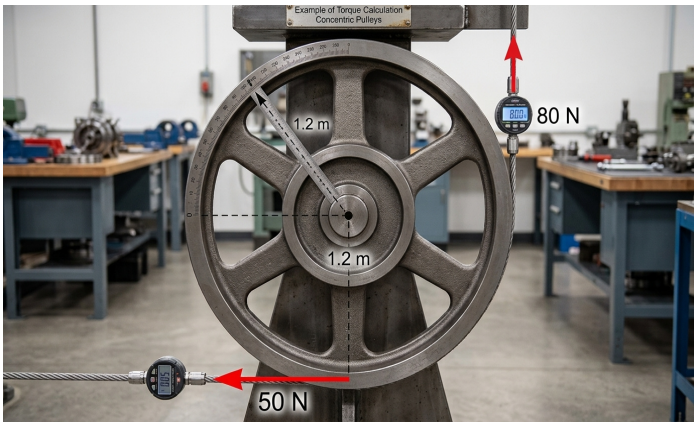
Direction of moment: *the moment can be clockwise or anti-clockwise direction*.

In two dimensions, the sense of a moment is described as either positive (anticlockwise) or negative (clockwise) as shown



Example: Calculate the moment of each force acting on the wrench and pulley as shown. Also tell direction of each moment.





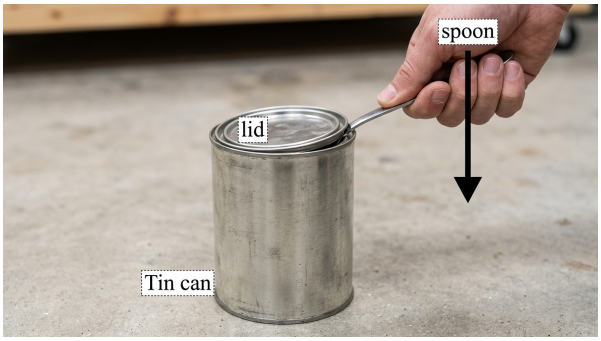
Daily Application of Moments

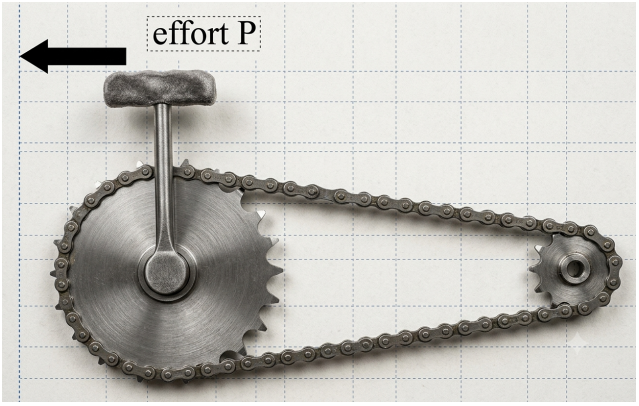
Draw on the following diagram the pivot, the force and the perpendicular distance.

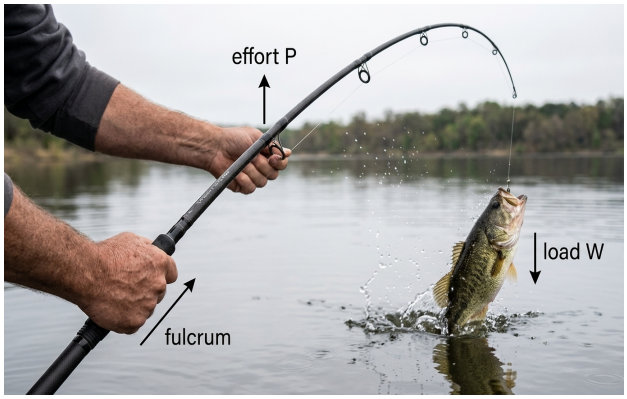
Indicate whether the direction of the moment too, clockwise or anticlockwise.



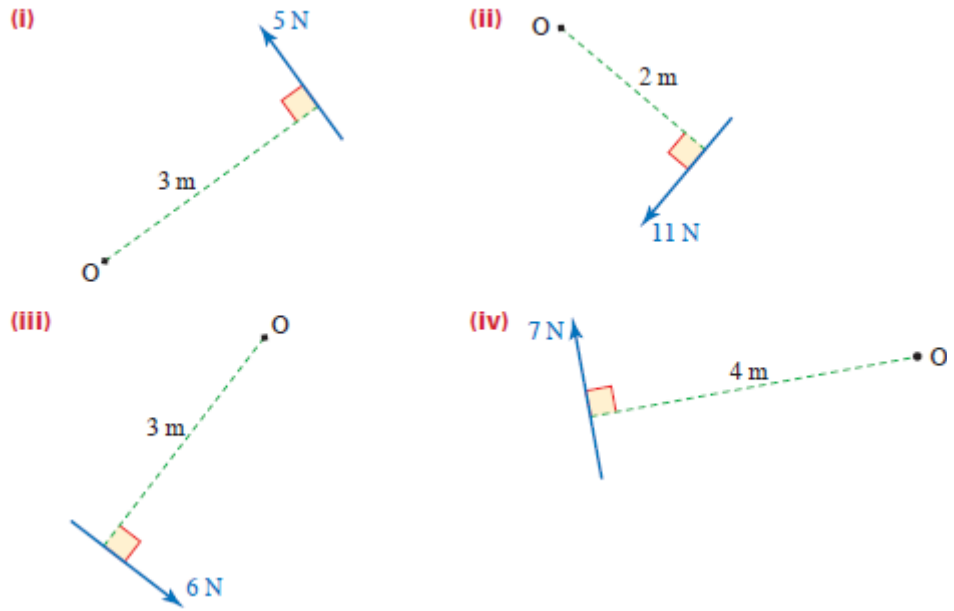
effort





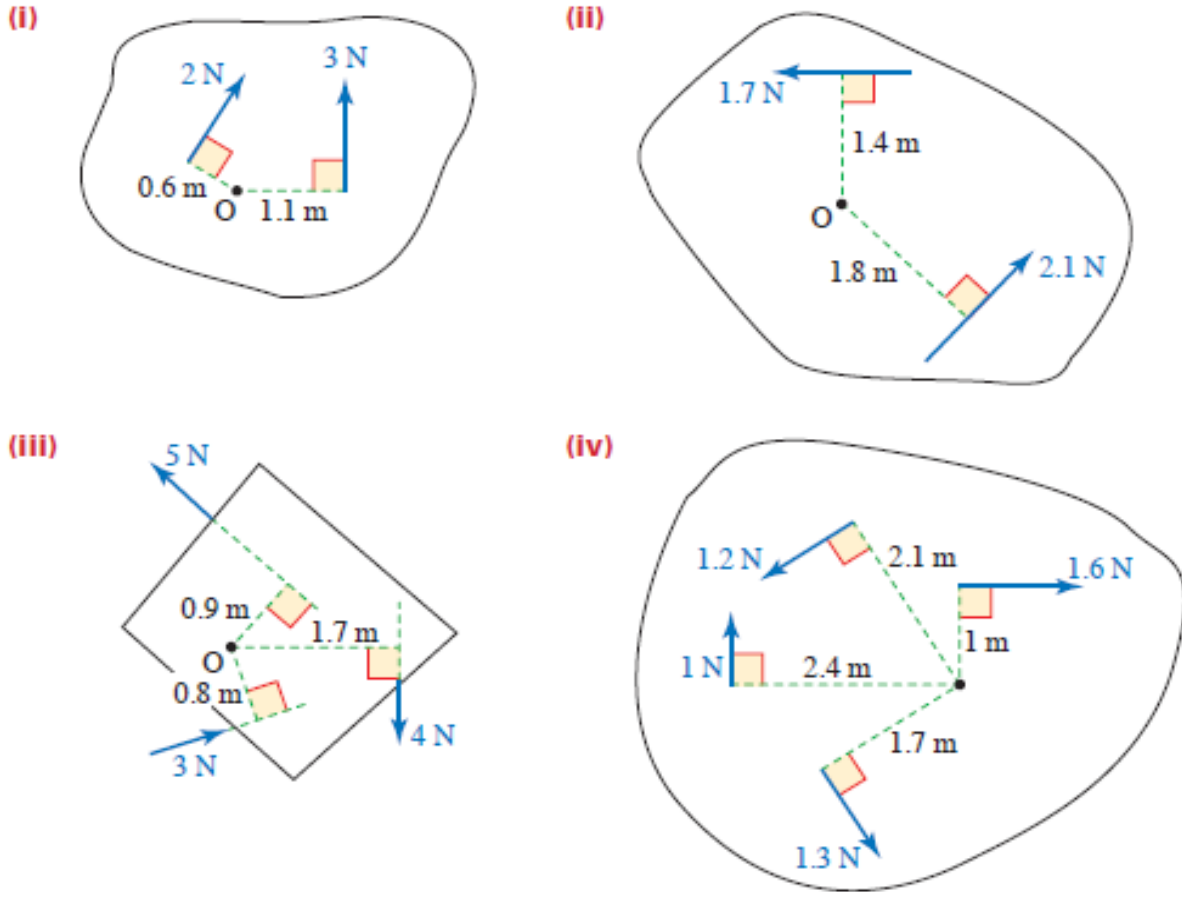


Example:In each of the situations shown below, find the moment of the force about the point and state whether it is positive (anticlockwise) or negative(clockwise).



Example:

The situations below involve several forces acting on each object. For each one, find the total moment.



4-Principle of Moment

Rotational equilibrium is obtained when the algebraic sum of the torques is **zero**.

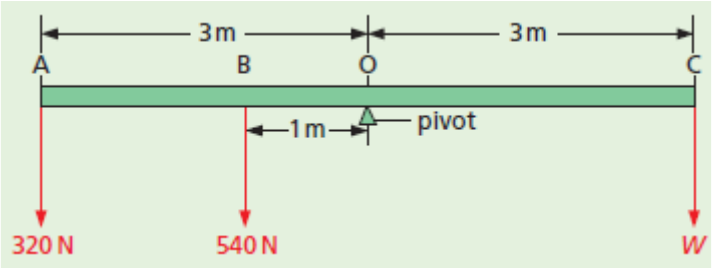
However, it can also be interpreted as:

Principle of Moment states that" *the sum of clockwise moment equals the sum of anti-clockwise moment for the body in equilibrium*".

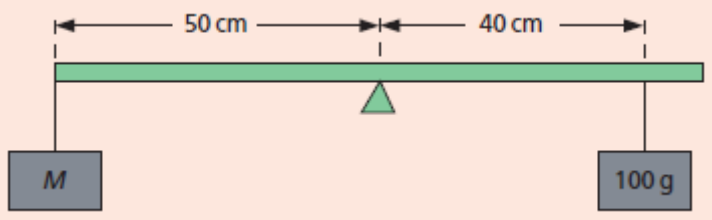
Sum of clockwise torques = sum of anti-clockwise torques

Example1:

The see-saw in Figure below, balances when Shani of weight 320 N is at A, Tom of weight 540 N is at B and Harry of weight W is at C. Find W.



Example2: The meter rule in Figure below is pivoted at its center. If it balances, find the mass M.



5-Parallel Forces

5.1 Two parallel forces can be replaced by a single force. The resultant of two parallel forces is given by: Direction: Same with the forces:

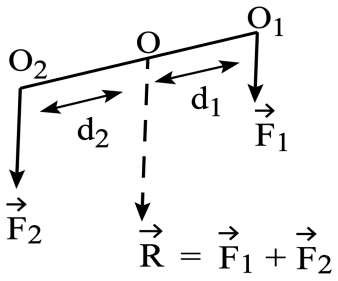
Location of resultant force: $\vec{R} = \vec{F}_1 + \vec{F}_2$

Clockwise moment: $M_0 = F_1 \times OO_1 = F_1 \times d_1$

Anti-Clockwise moment: $M_0 = F_2 \times OO = F_2 \times d_2$

Principle of moments, for the body in equilibrium, the sum of clockwise moments equals the sum of anti-clockwise moments.



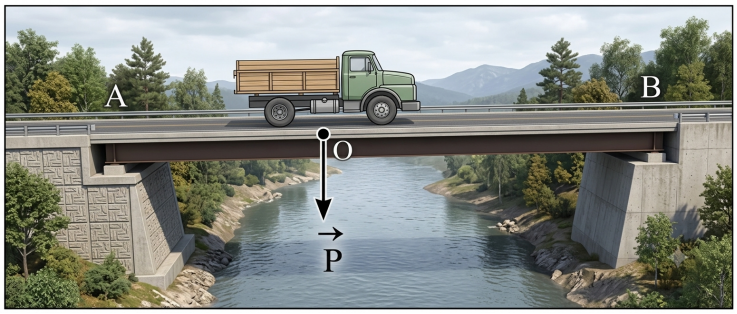


$$F_1 \times d_1 = F_2 \times d_2$$

*__Remark__: the sum of two forces F_1,F_2 equal the sum of tension forces in the string.

Example1: A uniform meter stick (100cm) is exerted by two parallel downward forces. The forces 20N and 90N exert at both ends. Find the magnitude and the location of resultant force.

5-2-Replacing a single force by two parallel forces



The weight of the car P can be replaced by two parallel forces on the supports.

Let O the application point of P and A,B application points of Q,F. The magnitude of resultant force:

P=F+Q

The two parallel forces are:

$$\frac{F}{OA}=\frac{Q}{OB}=\frac{F+Q}{OA+OB}=\frac{P}{AB}\Rightarrow F=\frac{P\times OA}{AB}$$

$$\frac{F}{OA}=\frac{Q}{OB}=\frac{F+Q}{OA+OB}=\frac{P}{AB}\Rightarrow Q=\frac{P\times OB}{AB}$$

Example: A 5m uniform beam is supported at the two ends (point A and B).The weight of 40 000N exerts on the beam at 2.6m from A. Replace the weight by two downward forces at A, B so that the beam is still in equilibrium. The weight of the beam can be neglected.

Solution:

$$\frac{F}{OA}=\frac{Q}{OB}=\frac{F+Q}{OA+OB}=\frac{P}{AB}\Rightarrow F=\frac{P\times OA}{AB}=\frac{40000\times 2.6}{5}=20800N$$

$$\frac{F}{OA}=\frac{Q}{OB}=\frac{F+Q}{OA+OB}=\frac{P}{AB}\Rightarrow Q=\frac{P\times OB}{AB}=\frac{40000\times 2.4}{5}=19200N$$

6-Center of Gravity

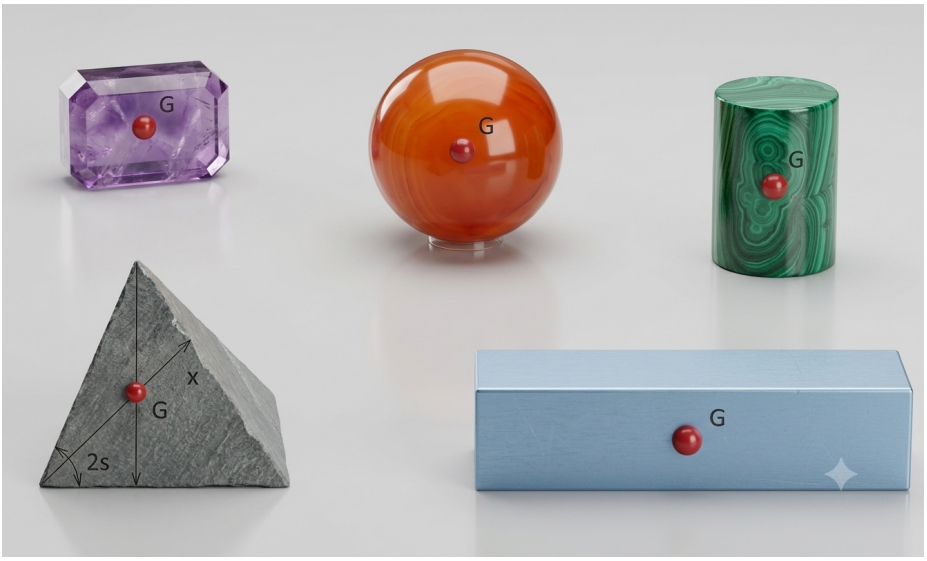


To design the structure for supporting a water tank, we will need to know the weights of the tank and water as well as the locations where the resultant forces representing these distributed loads act.

All objects are made of many particles. The weight of an object is due to the attraction of the Earth on all these particles. There is one point in any object where the whole weight seems to act.

Center of gravity is the point where the whole weight of the object acts or seems to act.

Note: Although the term center of gravity is widely used, the same imaginary point in a body may also be called the center of mass, since weight and mass are proportional in a uniform gravitational field. Because the center of mass does not require a gravitational field, many physicists prefer the term center of mass to center of gravity. Centre of gravity of common shapes:



7-Stability

Stability refers to the ability of a body to restore to its original static equilibrium, after it has been slightly displaced.

- A body in equilibrium experiences neither linear acceleration nor turning effect and, unless disturbed by an outside force, will continue in that condition indefinitely. (Newton's First Law)



☞ A body is in stable equilibrium if when slightly displaced and then released it returns to its previous position

Example: A hanging pendulum or a brick lying on a level plane.

☞body is **in unstable equilibrium** if it moves further away from its previous position when slightly displaced and released.

Example: A rule standing on its end or a ball standing on top of an inverted bowl.

☞*A body is in **neutral equilibrium** if it stays in its new position when displaced*

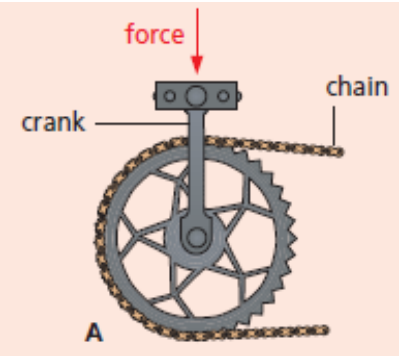
Example: A ball or a pencil lying on a level surface.

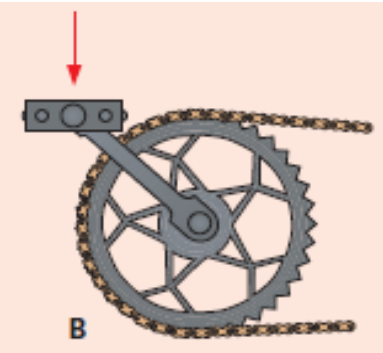
Stable bodies tend to have:

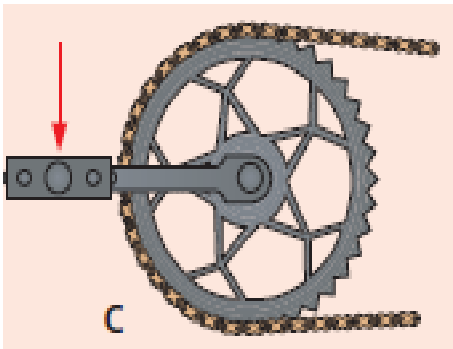
- Low center of gravity
- Wide base area

Practice Problems

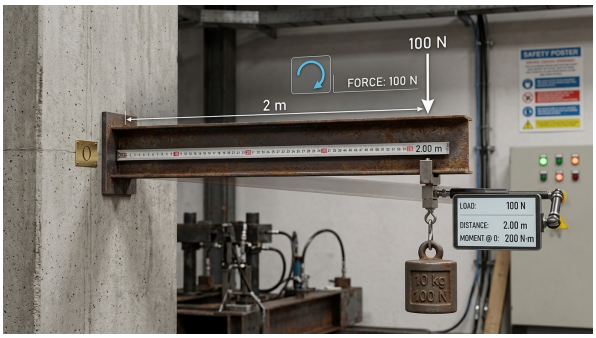
1- Figure below, shows three positions of the pedal on a bicycle which has a crank 0.20 m long. If the cyclist exerts the same vertically downward push of 25 N with his foot, in which case, A, B or C, has no turning effect?

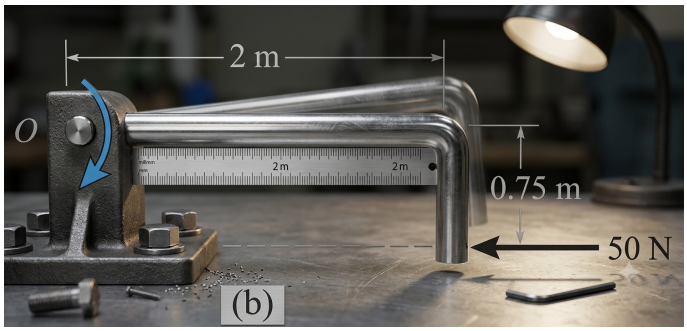




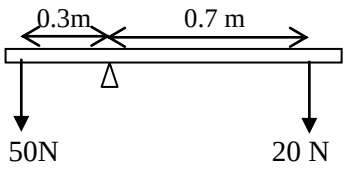
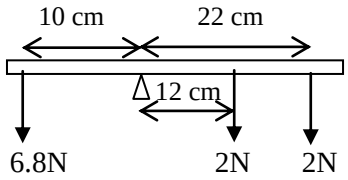


2-Find the moment of the following forces about the fix point 0.

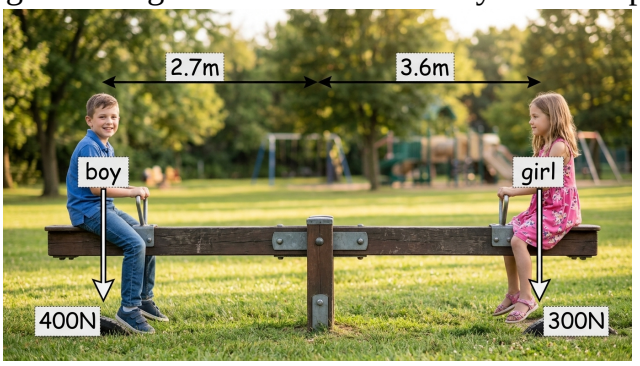




3-Calculate the total clockwise moment and the total anticlockwise moment in the following cases. Hence, determine if the rod will balance, turn clockwise or anticlockwise.

Diagram	Total clockwise moment	Total Anti-clockwise moment	Net Moment
			
			

4-The diagram represents a boy of weight 400N sitting on the left of a seesaw, 2.7m away from the pivot. A girl of weight 300N is sitting on the right hand side 3.6m away from the pivot.



1) Who is trying to turn the seesaw clockwise? _____

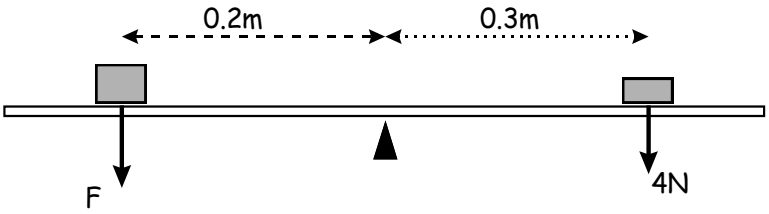
2) Who is trying to turn the seesaw anticlockwise? _____

3) What is the clockwise moment? _____

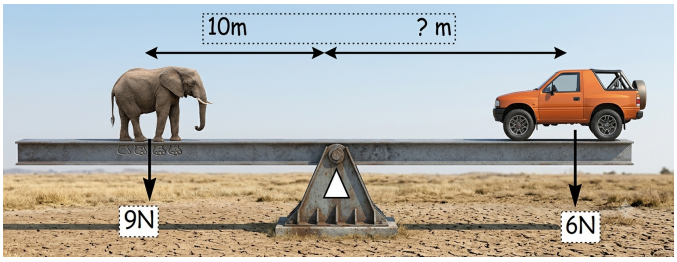
4) What is the anticlockwise moment? _____

5) Is the seesaw balanced? _____

5-The diagram below shows a ruler in equilibrium, What is the value of the force F which is balancing the 4N force?



6-The diagram below shows two toys being balanced by a young girl, find the distance of the truck from the pivot for the system to be balanced.



7- A 400 N child and a 300 N child sit on either end of a 2m long seesaw. Where along the seesaw should the pivot support be placed to ensure rotational equilibrium?

8- A uniform meter stick, supported at the 30cm mark, is balanced when a 0.50 N weight is hung at the 0.0 cm mark. What is the weight of the meter stick?

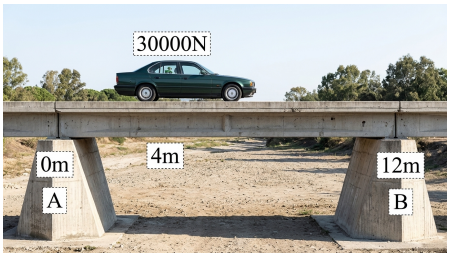
9- A pivot at the 35cm mark supports a meter stick. It is brought into balance by a 0.85 N weight suspended at the 5.5-cm mark. Compute the weight of the meter stick.

10- A uniform meter stick weighs 1.4N, and is supported by a pivot at the 65cm mark. Where must the 0.85 N force be suspended to balance the stick?

11-A 50cm uniform meter stick is suspended by two loads 600N and 400N at both ends. Find the resultant force and location of resultant force.

12-A 100cm meter rule is exerted by three forces with the same directions. The right end is exerted by 20N, the left end is exerted by 90N and at the middle is exerted by 30N. Find the net force.

13- A 12m long uniform bridge of negligible weights is being supported at the 2 ends. A car weighing 30000N parked at the 4m mark as shown. Replace the weight of car by two parallel force at A and B.



14-A 10m light beam is supported at the ends A and B. The weight of 40000N is placed at 4m from A. Find the downward forces exert on the supports A and B.